



**C23-AEI-BM-CH-CHOT-CHPP-
CHPC-CHST-EC-ECII– 301**

23101

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DAEI-THIRD SEMESTER EXAMINATION

ENGINEERING MATHEMATICS – II

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.

1. Evaluate : $\int (x^3 + x^2 + x) dx$

2. Evaluate : $\int \left(x + \frac{1}{x}\right)^2 dx$

3. Evaluate : $\int \frac{\sec^2(\log x)}{x} dx$

4. Evaluate : $\int (3x - 4) dx$

5. Evaluate : $\int_0^1 (x + 1) dx$

6. Evaluate : $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x dx$

7. Find the area of the region bounded by the curve $y = 3x^2$, the x -axis between the lines $x = 1$ and $x = 3$.

8. Find the mean value of $f(x) = x + x^2$ over the interval $[2,4]$.

9. Find the order and degree of the differential equation

$$\frac{d^3y}{dx^3} + 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} + 6x = 0$$

10. Find the Fourier coefficient a_0 for $f(x) = k$ in $0 < x < 2\pi$.

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

11. (a) Evaluate $\int \frac{1}{x^2 + 4x + 13} dx$

(b) Evaluate $\int \sin 7x \cos 3x dx$

12. (a) Evaluate $\int \frac{1}{(x+1)(x+2)} dx$

(b) Evaluate $\int x^2 e^{3x} dx$

13. (a) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx$

(b) Find the R.M.S. value of the function $f(x) = \sqrt{27 - 3x^2}$ over the interval $[0,1]$.

14. (a) Solve $\frac{dy}{dx} + \frac{y}{x} = x^2$

(b) Solve $\frac{dy}{dx} = \frac{1+x^2}{1+y^2}$

15. (a) Solve $(D^2 + 25)y = 0$.
- (b) Solve $(D^2 - 8D + 16)y = 0$.
16. (a) Evaluate $L\{e^{2t} + 2\sin 3t + 4\cos 3t\}$.
- (b) Evaluate $L\{t^2 e^{3t}\}$
17. (a) Evaluate $L^{-1}\left\{\frac{1}{s} + \frac{1}{s^2 + 9} + \frac{s}{s^2 + 9}\right\}$
- (b) Evaluate $L^{-1}\left\{\frac{s - 4}{s^2 - 8s + 25}\right\}$
18. Find the Fourier series for the function $f(x) = x$ in the interval $[0, 2\pi]$.

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